Georgios Anesiadis

Personal details

Name
Georgios Anesiadis

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Thermopylon 33
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GitHub
github.com/GeorgeAnes

LinkedIn
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Core skills

Python
SQL
MATLAB
pandas / NumPy
scikit-learn / PyTorch
FastAPI / React
LangChain / LangGraph
Local LLMs
Process mining
Power BI
Git

Languages

English	C2
German	C1
Greek	Native
Dutch	A1

Profile

MSc Artificial Intelligence & Engineering Systems student at TU/e with a chemical engineering background and hands-on experience in engineering analytics, GenAI workflows, process mining, and industrial data modelling. Built FastAPI/React and Python-based tools for document risk auditing, event-log KPI analysis, FEM simulation analytics, and predictive control. Interested in AI & Data / technology consulting roles where technical depth supports measurable business and engineering decisions.

Education

MSc Artificial Intelligence & Engineering Systems — **Sept 2025 - Present**
Eindhoven University of Technology (TU/e), Eindhoven, NL
- Track: High Tech Systems & Robotics; current GPA 8.7/10.
- Focus: AI for engineering systems, computer vision, optimization, embedded AI, and data-driven modelling.

MEng Chemical Engineering, Integrated Master's (300 ECTS) — **Sept 2019 - Jul 2025**
Aristotle University of Thessaloniki, Thessaloniki, GR
- GPA 8.54/10; valedictorian, ranked 1st in class.
- Thesis: dynamic modelling of solid-liquid separation in a decanter centrifuge using gPROMS.

Employment

Research Intern - Engineering Analytics & Agentic Design — **Sept 2025 - Present**
ASML / TU Eindhoven, Eindhoven, NL
- Design and implement a local AI-assisted design exploration framework for electromagnetic actuators, combining FEM simulation outputs, KPI-driven evaluation, optimization tools, and LangGraph agent orchestration.
- Built Python workflows to clean, validate, and analyze actuator simulation data, translating complex parameter-KPI behavior into design trade-off insights for TU/e and ASML engineering stakeholders.
- Implemented a surrogate-assisted Pareto optimization pipeline to explore actuator design regions, filter infeasible candidates, cluster optimal solutions, and support engineer-facing recommendations.

Industry Research Collaborator - Process & Data Modelling — **Apr 2025 - Apr 2026**
Danone / Aristotle University of Thessaloniki, Eindhoven & Thessaloniki
- Validated and extended a physics-based decanter centrifuge model using lab, pilot, semi-industrial, and plant-scale data; compared model predictions against experimental and industrial measurements.
- Translated parameter-estimation and model-calibration results into insights on process performance, scale-up robustness, operational constraints, and manufacturability.

Research Intern - Catalysis & Sustainable Energy — **Jun 2023 - Sept 2023**
CERTH, Thessaloniki, GR
- Synthesized and characterized catalysts for sustainable-energy and environmental applications; analyzed laboratory results and documented findings for internal technical transfer.

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Technical stack

AI / Data
Python; pandas; NumPy; scikit-learn; PyTorch; SQL

Agentic AI
LangChain; LangGraph; local LLMs; RAG; tool orchestration

ML Analytics
feature engineering; model evaluation; surrogate modelling; KPI design

Product Stack
FastAPI; React; API-driven dashboards; Git; reproducible workflows

Process Analytics
process mining; bottleneck analysis; SLA monitoring; reporting automation

Engineering
ANSYS Maxwell; gPROMS; GAMS; Honeywell UniSim; MATLAB

Selected Projects

Enterprise AI Document Risk Auditor
Built a local FastAPI/React application auditing business documents for unsupported claims and grounding risk. Audits claims, retrieves evidence, exports reports, and adds optional LM Studio/Gemma reviewer notes while preserving an auditable baseline.

Process Mining KPI Dashboard
Developed a retro enterprise operations dashboard for IT service event logs. Surfaces throughput, SLA leakage, bottlenecks, variants, rework, and executive recommendations through a multi-window UI.

Facial Expression Recognition with Classical ML
Built a computer-vision pipeline for laptop-camera emotion recognition using face detection, HOG features, SVM/MLP/Random Forest classifiers, and confusion-matrix evaluation on public/private splits.

Petri-Net Discovery with Genetic Algorithms
Implemented a DEAP-based optimization pipeline to discover Petri-net structures from event-log traces, comparing evolutionary operators and fitness strategies for process-mining workflows.

AERO MPC / SPC / Koopman Control Benchmark
Compared model-based and data-driven predictive controllers for a pitch/yaw AERO system in MATLAB, including linear MPC, nonlinear MPC, SPC, and Koopman-lifted SPC.

Publication

First-author paper accepted in Chemical Engineering Science **2026**
An efficient modelling and simulation framework of a decanter centrifuge for Solid-Liquid Separations. DOI: 10.1016/j.ces.2026.123655.

Leadership & Activities

Vice President for Human Resources / Board Member, BEST Thessaloniki **Aug 2022 - Aug 2023**
Led recruitment, onboarding, and internal training for a 15-member HR team in a European student engineering network.

Awards & Distinctions

Outstanding Performance Award, ISEC-3

Best Delegate, Model United Nations

Erasmus+ exchanges

NASA Space Apps & Copernicus Hackathons